Unit 6

Data Communication and Computer Network 5 hours

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6.1 Data Communication

6.1.1 Introduction

Data communication technologies deals with means and method of data transfer from one location to another. Marriage between computing and data communication technologies is one of the most exciting developments that has revolutionized information age. This development gave birth to computer networks.

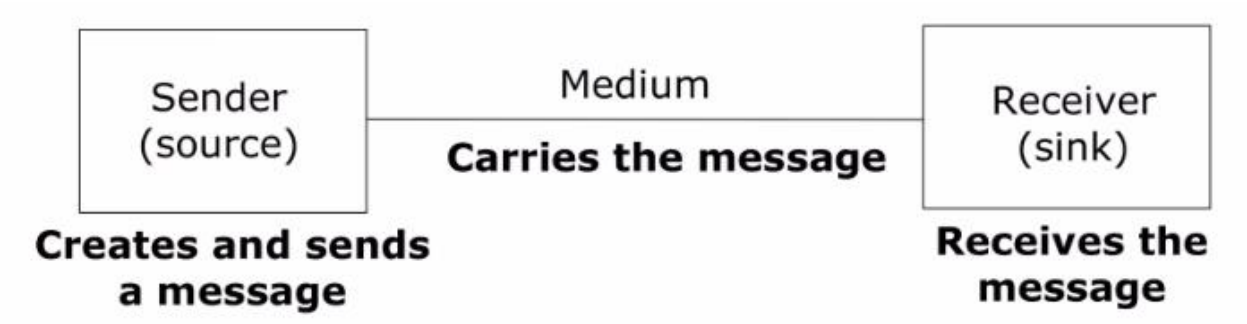
A computer network is a network of computer that are geographically distributed, but connected in a manner to enable meaningful transmission and exchange of data among them. Sharing of information, resources (both hardware and software), and processing load is the main objective of a computer network.

Basic Elements of a Communication System

Communication is the process of transferring a message from one point to another. Three basic elements of any communication system are as:

1. A sender (source) that creates and sends a message.
2. A medium that carries the message.
3. A receiver (sink) that receives the message

For example, when you speak to your friend on telephone, you are the sender, telephone line through which your voice is transmitted is the medium, and your friend is the receiver. This is the simple example of voice communication. In case of data communication, sender and receiver are normally machine, in particular, computer and associated devices (computers, terminals, peripheral devices like printer, plotter, disk, etc.), and transmission medium my be telephone lines, microware links, satellite links, etc. Hence electronic systems used to transfer data from one point to another point are called data communication systems. Unlike computer that process data, data communication systems transmit data from one point to another.



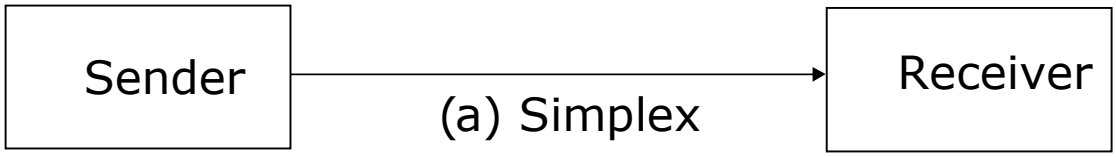
Data Transmission models

The three models of data transmission are simplex, half-duplex, and full-duplex.

1. **Simplex:** A simplex communication system can transmit data in one direction only. Devices connected to such a circuit are either send-only (such as keyboard) or receive-only (such as printer).

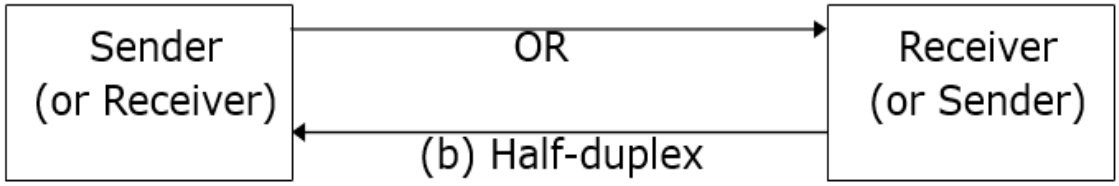
   At first through, this mode may appear adequate for applications in which flow of instruction

is unidirectional. However, in almost all data processing applications, bidirectional communication is required because even when flow of data is unidirectional (such as from a terminal to a computer), a return path is generally needed to send acknowledgement, control, or error signals. Without this capability, a sender might send data and never know that the receiver never received it (due to some problems somewhere). Hence, simplex circuits are seldom used.



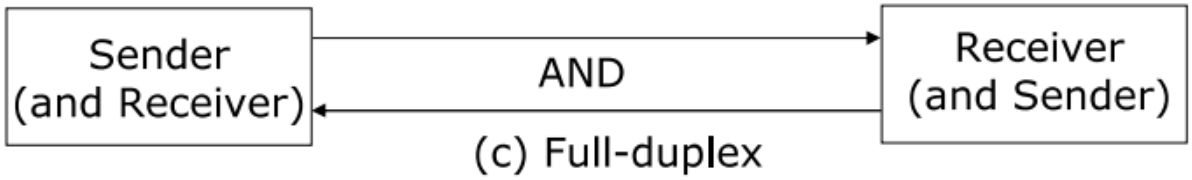
Examples of Simplex mode are radio, keyboard, and monitor.

2. **Hal-duplex:** A half-duplex communication system can transmit data in both directions, but in only one direction at a time. Hence id can alternately send and receive data. It required two wires. It is most suitable for voice communication using telephone in which one person speaks at a time. It is suitable also for connecting a terminal to a computer in which the terminal transmits data and then the computer responds with acknowledgement.



Example of half-duplex is Walkie-Talkies.

3. **Full-Duplex:** A Half-duplex system needs to switch direction each time the direction of data transfer receives. This requires a special switching circuits and a delay about 150miliseconds. When compared with the high processing capabilities of modern computers, this delay is often unacceptable. Moreover, some applications require bi-directional data transfer simultaneously. In such cases, a full duplex system is used that allow data of flow in both the directions simultaneously. It requires four wires. It improves efficiency because it eliminates the bi-direction switching delay of a half-duplex system.



Example of the Full-duplex mode is a telephone network.

Data Transmission Speed

Bandwidth refers to data transfer rate of a communication system (amount of data that it can transfer per unit of time). It is analogous to a road's width. Wider a road, the more traffic it can handle in a given time. Similarly, higher the bandwidth of a communication system, the more data it can transfer in a given time.

Bandwidth is measured in bit per seconds (bps) (also called as baud). Generally, baud is identical to bits per seconds (bps). Hence a rate of 300 baud means 300bps. However, technically baud refers to number of signal (state) changes per seconds. Hence, using more sophisticated coding techniques 1baud can represent 2 or even 3bps. However, for most communication system, 1 baud represents one signal changes per seconds, and is equivalent to 1bps.

Based on data transmission speeds, three categories of communication channels (paths) are:

1. **Narrowband**

   Narrowband or sub-voice grade channels have speed range of 45 to 300baud. They are used to handle low data volumes, and are adequate for low-speed devices. They are used for telegraph lines and low speed terminals.

2. **Voiceband**

   Voiceband channels have speed up to 9600 baud. They are so called because their major application is in ordinary telephone voice communication. They are also used for data transmission from slow I/O devise to CUP and vice versa.

3. **Broadband**

   Broadband channels are used for transmission of large volumes of data at high speed. They have speed of 1 million baud or more. Broadband facility is used for high-speed computer to computer communication or for data transmission to several different devices simultaneously.

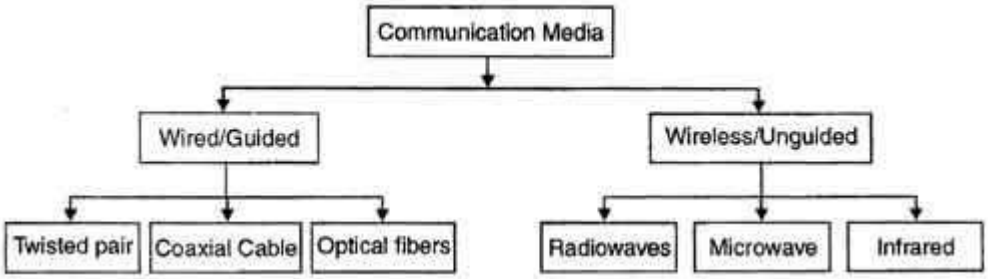
Data Transmission Media

Communication media refers to the means (transmission and storage tools or channels for data storage and transmission) of delivering and receiving data or information. Data communications can be carried from one location to another through a variety of communications channels. These communications media include telephone lines(twisted pairs), coaxial cable, fiber-optic cable, microwave, and satellite. In general, there can be two ways of connecting microcomputers with each other and with other equipment: guided and radiated media.

The Communication Medium plays an important role in Networks. If the medium works well and properly, then the speed of transferring data is good but if the medium is not working properly, then the data would be delayed or would not be sent or even can be lost during transmission.

- **Guided Media:** It refers to channels that allow the transmission of data and information through a physical media such as a twisted pair wire, coaxial cable, or fiber optic cable.

- **Unguided (Radiated) media:** Unguided media refers to those that transmit data and information through the air such as microwave, or satellite.



Twisted Pair Wire

- Most popularly used communication media
- Twisted pair cable consists of a pair of insulated wires twisted around each other to minimize interference from others i.e. the twisting helps to reduce noise pickup from outside sources and crosstalk on multi-pair cables.
- It is cheapest and easily available wire, which can carry data to large distance without data loss.
- Mostly used in telephone systems to carry most of the voice and data communication.
- It has fewer bandwidths as compared to other wires.
- They are available in varieties of grades called levels.
- There are two types of twisted pair cables

a) ***Shielded twisted pair (STP):*** STP simply has additional shielding material that is used to cancel any external interference that may be introduced at any point in the path of the cable. STP is used in noisy environments where the shield around each of the wire pairs, plus an overall shield, protects against excessive electromagnetic interference. Using STP cables ensure that we get the maximum bandwidth from your cablingeven if the external condition is less than ideal. The biggest drawback to using STP cables is the higher cost.

b) ***Unshielded twisted pair (UTP):*** UTP cables have no protection against such interference and its performance is often degraded in its presence.

Coaxial Cable

- It consists of a single thick copper wire in the core, surrounded by an insulation plastic with a net of thincopper wires used as earthing and in the outermost part- it is insulated by a thick plastic
- It generally comes into two categories: thin coax and thick coax
- Thick coax generally has 1 cm core diameter and has high degree of noise immunity and high bandwidth. Ithas 100base5 (100 mbps bandwidth with 500m wire length)
- Thin coax has half of the size and more than half of the bandwidth in comparison to thick. It has 10base2bandwidth.
- Relatively expensive than twisted pair cable.
- Because coaxial cables have very little distortion and are less prone to interference, they have low error rates.
- It is used in TV in our homes, as well as computers in local area networks are linked by coaxial cables.

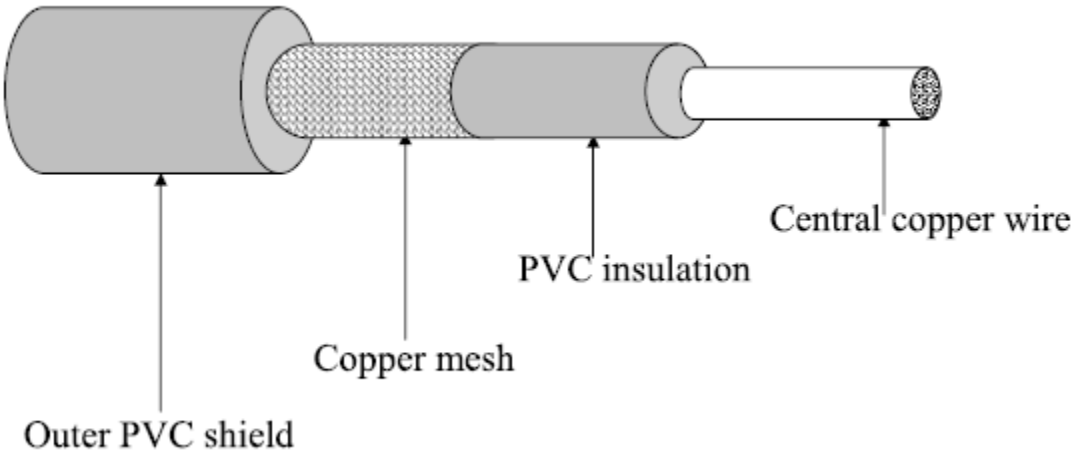


Fig: Coaxial Cable

Optical fibers / Fiber-Optic Cable

- It is the most advanced media in communication, which uses light rather than electricity to transmitinformation.
- It is a flexible, thin, transparent fiber made of high-quality glass or plastic and is very hard to identify with ournaked eyes.
- It has two layers: clad and core
  Both clad and cores are made up of same material (plastic or glass). The light is passed through an opticalfiber by the process of total internal reflection. Light source for its „laser‟ or „led‟ and detector is „pin‟. At thesource, electrical pulses generated by computer or electronic devices, is converted into light by „laser‟ or „led‟which flows form optical fiber in the form of light. At the destination, the light rays are again converted into electrical pulses by „pin‟.

- It doesn't have any danger of leakage and electrical interferences. So it can be used in any environments likeunder big factories, ground or airports i.e. it has high level of security.
- It has highest bandwidth and highest speed in comparison to any other Medias.
- It is most expensive
- Data can travel long distance without any loss and amplification, so amplification cost can be minimized byusing optical fiber.
- Fiber optics can generate high Data Rates, so these are used for long distance communications that requirehigh speed and least data loss.

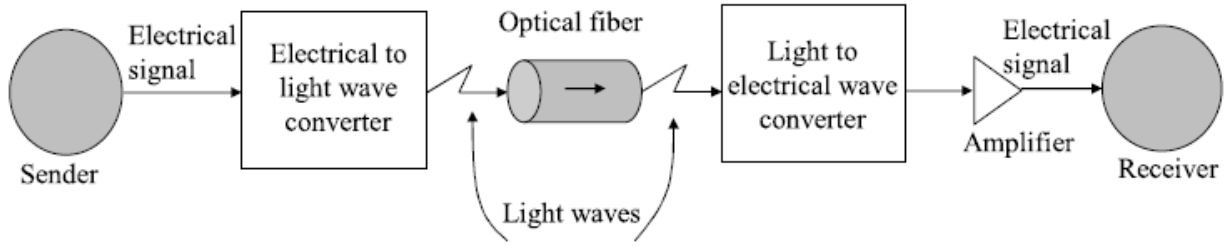


Fig: Optical Fiber

Microwave System

Microwave data transmission differs from the other communications channels in that data is transmitted through the air instead of through cables or wires. Microwaves are high-frequency radio waves that can only be directed instraight lines. Consequently, microwave transmission is usually limited to communications occurring within the limits of a particular city or community. For microwave transmissions to be able to occur over larger distances, data messages must be relayed from one location to other using antennas placed at high altitudes usually 20-30 miles apart.

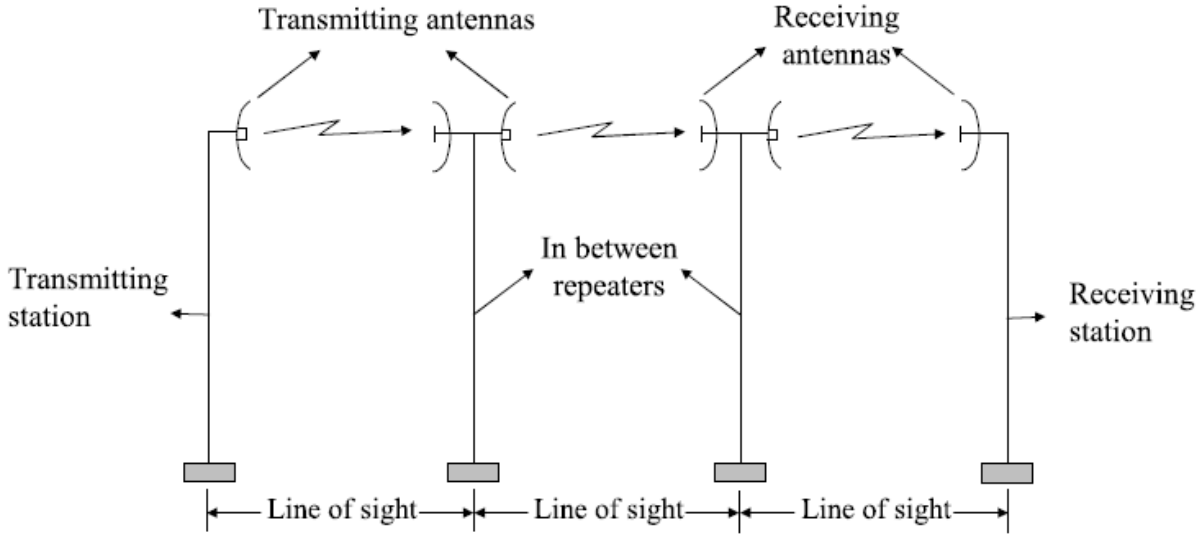


Fig: Microware Communication System

Communication Satellites

Instead of antennas, satellites can also be used to transfer microwave messages from one location to another. Satellites rotate approximately 23,300 miles above the earth in precise locations. Satellite transmission stations that can both send and receive messages are known as earth stations.

This is a space station that receives microwave signals from an earth-based station, amplifies the signals, and broadcasts the signal back over a wide area to any number of earth-based stations. A transmission from the earthto a satellite is called an **uplink;** a transmission from a satellite to an earth station is called a **downlink.** Communication satellites are used in application such as air navigation, television and radio broadcast, videoconferencing and paging.

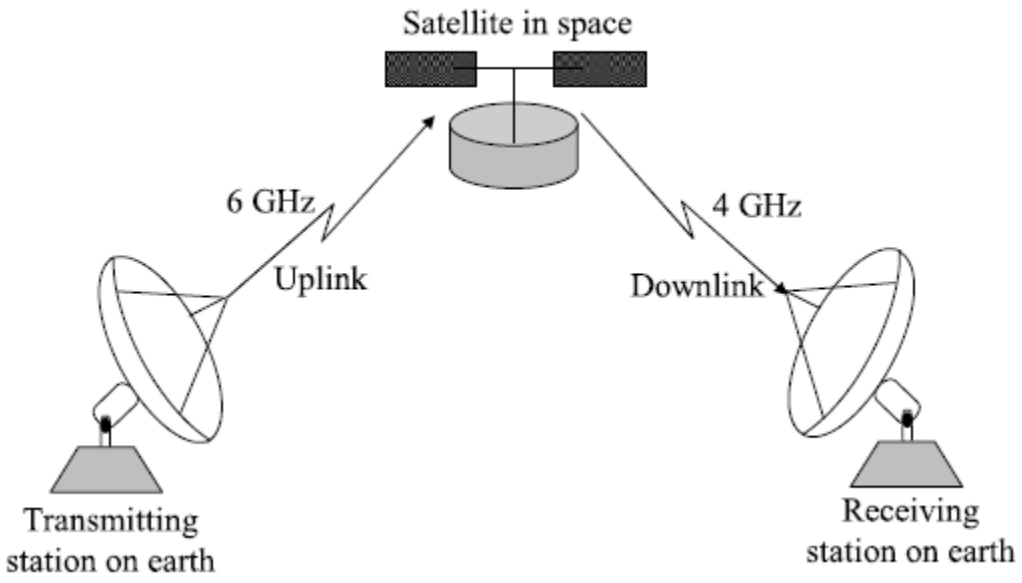


Fig: Satellite communication system

Advantages:

- A major advantage of satellite transmission is that large volumes of data can be communicated at once.
- Cost of satellite capacity does not increase with the number of users/receive sites, or with the distancebetween communication points.
- Satellite communications can operate independently from terrestrial infrastructure.
- Broadcasting of information's is possible.
- Provision of service to remote or underdeveloped areas

Disadvantages:

- The major drawback is bad weather can severely affect the quality of satellite transmissions.

- It has a serious security problem, because it is easy to intercept the transmission as it travels through theair.
- Congestion of frequencies and orbit

6.2 Computer Network

Definition

Computer network is a collection of computes and devices interconnected by means of communication channels that provides communication among users and allows users to share resources. A **computer network** is the infrastructure that allows two or more computers (called **hosts**) to communicate with each other. The network achieves this by providing a set of rules for communication, called **protocols**, which should be observed by all participating hosts

In 1960s the advanced research project agency (ARPA) started funding the design of the Advanced Research Project Agency Network (ARPANET) for the United States department of defense. It was the first computer network in the world, the development of the network began in 1969.

Computer network can be used for a variety of purpose and some of them are:

- Sharing of hardware, software's
- Sharing of files, data and information's
- Information preservation
- Security
- Speed up communications etc.

Advantages

a) **Sharing of resources**

Computer consists of various kinds of resources such as files, data, software's or hardware's. Computer network provides an efficient mechanism to share such resources among the users at thesame time without intervening each other's.

b) Increased storage capacity

There are multiple computers connected in a network where each computer can easily share files stored in a computer. A standalone computer might fall short of storage device but when many computes in a network, memory of all computers can be used

for storage. We can also design storage servers on a network in order to have huge storage capacity.

c) Increased cost efficiency

There are varieties of soft wares are available in the market which are costly and take time for installation. Computer network resolves this issue, as the software can be stored or installed on a system or a server and can be used by the different workstations.

d) Backup and recovery

Generally, a powerful computer is used as server is placed in a secured place and good mechanism is also provided for backup of the data. If the data is lost or damaged in the clients (other workstations), then it is possible to restore those data from the server.

e) Cheaper and faster communication

Data, information's and messages can be passed from one computer to another computer within few seconds. If the message is to be passed to a long distance, then it saves time as well as cost of the message transfer.

f) Centralized control

All the computers are connected to a main server. There are many essential networking policies, rules or security issues for a successful networking that must be implemented, which are applied in the main server. Such rules are automatically transferred or applied in the clients connected to the servers i.e. all the controlling issue of all the computers and resources is possible form the server.

Disadvantages

a) Security issue

It is the main drawback of computer network. If a computer is connected on a network, a computerhacker can get unauthorized access by using different tools. In case of big organizations, different network-security software's are used to prevent the theft of any confidential and classified data. However, they are not efficient to prevent data.

b) Rapid spread of computer viruses

If one computer in a network is affected by computer viruses, there is possible threat of other systems getting affected too. Viruses get spread on a network easily because of the interconnectivity of workstations. Such spread can be dangerous if the computers have important database which can be corrupted by the viruses.

c) Expensive setup

The initial set up cost of a computer network can be high depending on the numbers of computers to be connected. An organization have to buy NIC (Network Interface Card) for each workstation, in case they are not in-built, various networking devices such as routers, hubs, switches, cables etc. which are costly and are required for installation of computer network.

d) Dependency on main server

All the workstations are connected in a networking are connected through a main server. If the server breaks down, the whole network system is affected and the network system becomes useless. In big organizations a powerful and highly secured computers are used as a main server.

e) Needs special technical knowledge

Network administrator, a highly skilled manpower, is required to set up, operate and maintain anetwork. The under developed countries like Nepal, it is also a major problem.

Types of Computer Network (PAN, LAN, CAN, MAN and WAN)

According to size of network, speed of data transmission, distance covered by the computers, technology used, structure of network, etc computer network is classified in to three groups: LAN, MAN and WAN, however networks are often classified by their physical or organizational extent or their purpose.

Personal Area Network (PAN)

A Personal Area Network is typically a short-range communication network among computerized devices such as telephones, fax machines and digital media adapters. The PAN may extend only a few yards, it can be used to link to other networks such as the Internet.

PAN - is a computer network organized around an individual person. Personal area networks typically involve a mobile computer, a cell phone and/or a handheld computing device such as a PDA. We can use these networks to transfer files including email and calendar appointments, digital photos and music.

A **personal area network** (PAN) refers to the interconnection of information technology devices or gadgets within the environment of an individual user (typically within 10 meters or 33 feet). These interconnected devices might include laptop computers, PDAs, cell phones, printers, PCs or other wearable computer devices. It is also known as a **wireless personal network (WPAN)**.

College (Campus) Area Network (CAN)

A campus area network (CAN) is a computer network made up of an interconnection of LANs within a limited geographical area. The networking equipment (switches, routers) and transmission media (optical fiber, copper plant, Cat5 cabling etc.) are almost entirely owned by the campus tenant / owner: an enterprise, university, government etc.

In the case of a university campus-based campus network, the network is likely to link a variety of campus buildings including, for example, academic colleges or departments, the university library, and student residence halls.

Local Area Network (LAN)

A local area network (LAN) is a network that connects computers and devices in a limited geographical area such as home, school, computer laboratory, office building, or closely positioned group of buildings i.e. A LAN is a computer network limited to a small area such as an office building, university, or even aresidential home i.e. it is a network within a very limited area extended up to few kilo meters. Most mid tolarge-sized businesses today use LANs, which makes it easy for employees to share information.

Characteristics of LAN

- It is simpler and cheaper than others.
- Normally it is owned by a private organization or an individual
- The data transmission rate is normally 1 to 100 mbps.
- Error rates are much lower in comparison to others. Generally, it is 100 times lower than in WAN.
- Generally guided media such as twisted pair cables, optical fibers etc are used for datatransmission
- Same kind of protocol is used.

- It is highly secured because no one unauthorized users can access the resources of a privatenetwork.

Metropolitan Area Network (MAN)

A Metropolitan Area Network (MAN) is a network that connects two or more computers, communicating devices or networks in a single network that has geographic area larger than that covered by even a large 'Local Area Network' but smaller than the region covered by a 'Wide Area Network'. MANs are mostly built for cities or towns to provide a high data connection and usually owned by a single large organization. A Metropolitan Area Networks bridges a number of 'Local Area Networks' with a fiber-optical links which act as a backbone, and provides services similar to what Internet Service Provider (ISP) provide to Wide Area Networks and the Internet.

Characteristics of MAN

- It is extended up to a city or large geographical area.
- Many heterogeneous systems can be interconnected.
- Different communication media: guided or unguided media, can be used
- Different types of communication protocol can be used.
- It has high chance of data leakage or security break.

Wide Area Network (WAN)

A wide area network (WAN) is a computer network that covers a large geographic area such as a city, country, or spans even intercontinental distances, using a communications channel that combines many types of media such as telephone lines, cables, and air waves. It is also known as ***Long-Haul network.*** A WAN often uses transmission facilities provided by common carriers, such as telephone companies. WAN technologies generally function at the lower three layers of the OSI reference model: the physical layer, the data link layer, and the network layer.

Characteristics of WAN

- It is extended up to a large area, even throughout world
- It is combination of different types of LAN and MAN.
- Generally owned by multiple organizations.
- It has massive amount of heterogeneous systems.

- Data transmission rate is slow, normally less than 1mbps.
- Its operation cost is very high
- Communication is done through unguided media such as micro waves, satellites. etc
- Error rate is also high.
- E.g. Internet

Comparison among 5 computer networks:

S.N.	Features	PAN	CAN / LAN	MAN	WAN
1	Geographical distribution of nodes	Within few meters	With few kilometers	Within in few tens of kilometers (City)	May extend over several thousand kilometers
2	Purpose	To enable communication among computing devices within an office room or house of an individuals	To enable communication among computing devices within a building (LAN) or a campus (CAN) of an organization	To interconnect LANs located in a city or metropolitan	To interconnect connect LANs spread across cities or countries
3	Communication Channel used	Infrared, radio wave	Twisted pair, coaxial cable, fiber optics	Fiber optics, telephone lines, microwave links	Fiber optics, telephone lines, microwave links, satellite channels
4	Ownership	Personal	Single organization	Single or Multiple	Usually Multiple
5	Communication cost	Negligible	Low	Medium	High

Networkk Topologies

A network topology is thelayout of the interconnections of the nodes of a computer network. Computer topology is the logical way of explaining how computers are connected to each other and shows physically how they are linked. The major goal of topology is to find out the most economical and efficient way to connect all the users to the network.

Network topologies may be physical or logical. Physical topology means the physical design of a networkincluding the devices, location and cable installation. Logical topology refers to how data is actually transferred in a network as opposed to its physical design. In general, physical topology relates to a core network whereas logical topology relates to basic network.

BUS TOPOLOGY

In this structure all the computers are attached to a single cable called **bus**. So all the nodes shares the bus,the server is located either side and terminator is located on other side of the bus to indicate that the networking is not available beyond that point. It is based on broadcasting architecture.

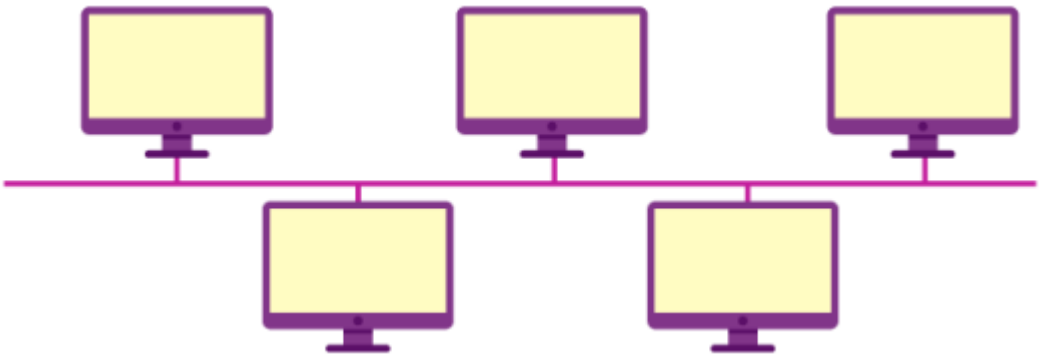


Fig: Bus topology

Advantage:

- If any one of the compute in network is down, then whole system is not affected.
- It is more flexible, we can connect any number of computers by using connector in the bus.
- It helps in reducing the number of physical lines.

Disadvantage:

- Data traffic is high, because all the data aretransmitted from the bus.
- Data collision is high.

Fig: Bus Topology

- If the shared communication channel (bus) fails, entire network fails.

STAR TOPOLOGY

A central connecting device called hub connects all the nodes in the network, which is connected to a powerful central computer called server i.e. it connects its

nodes to each other through a host node, and nodes can communicate via the host node only. It is also based on broadcasting architecture. The hub acts as a signal booster or repeater. The star topology is considered the easiesttopology to design and implement.

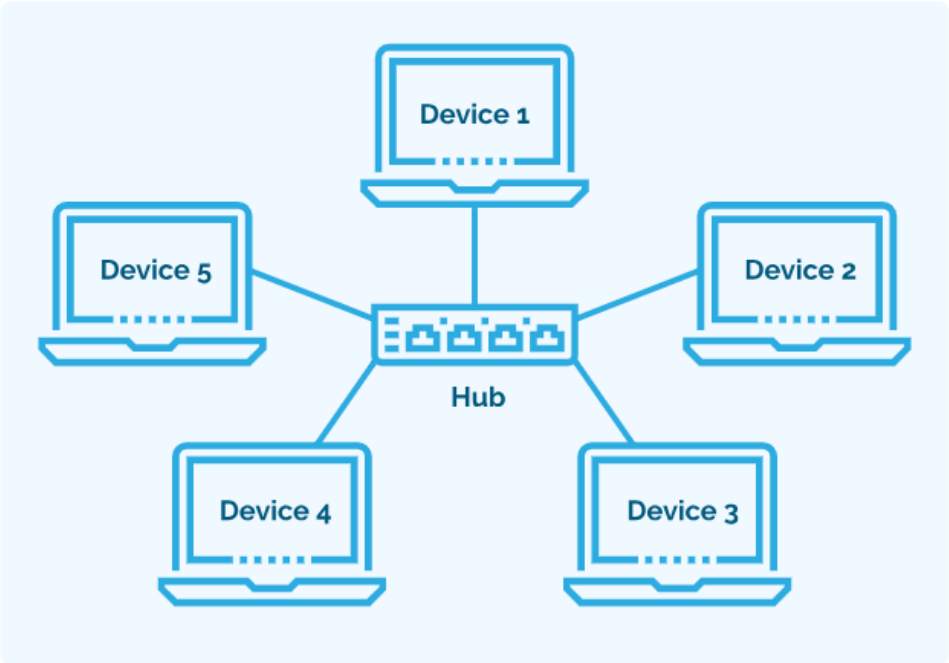


Fig: Star topology

Advantage:

- Easy to diagnose fault because of hub.
- If any of the computers in the network goes down, then the other computers can carry out theirfunctions.

- Transmission delays between two nodes do not increase by adding new nodes to a network,because only two links connect any two nodes.

Disadvantage:

- The system crucially depends on host node (hub), if it goes down the whole system fails.
- Cost of cable is more.
- At the time of broadcasting, all the nodes remain busy.

RING TOPOLOGY

Each computer is connected in a circular structure. Each computer is connected to other two computers in either side. It is a network topology that is set up in a circular fashion in which data travels around the ringin one direction. It is based on peer-to-peer architecture.

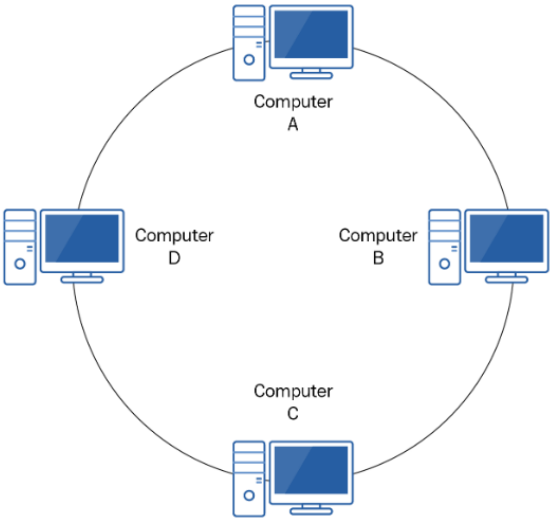


Fig: Ring topology

Advantage:

- All the computers are independent. So each computer acts as client as well as server.
- Since data travels in one direction, there is less chance of collision and less data traffic.
- Every computer has equal access facility to the resources.
- If link between two nodes fails, or if one of the nodes fails, alternate routing is possible.

Disadvantage:

- Non-flexible.
- Addition of new nodes in a network increases communication delay because communication delayis directly proportional to number of nodes in the network.

TREE TOPOLOGY

It is also known as hierarchy network. In this structure, a central node, called root is connected to one or more nodes (second level) with a point-to-point link. The second level nodes will also have one or more other nodes that are one level lower in hierarchy connected to it, also with point-to-point link.

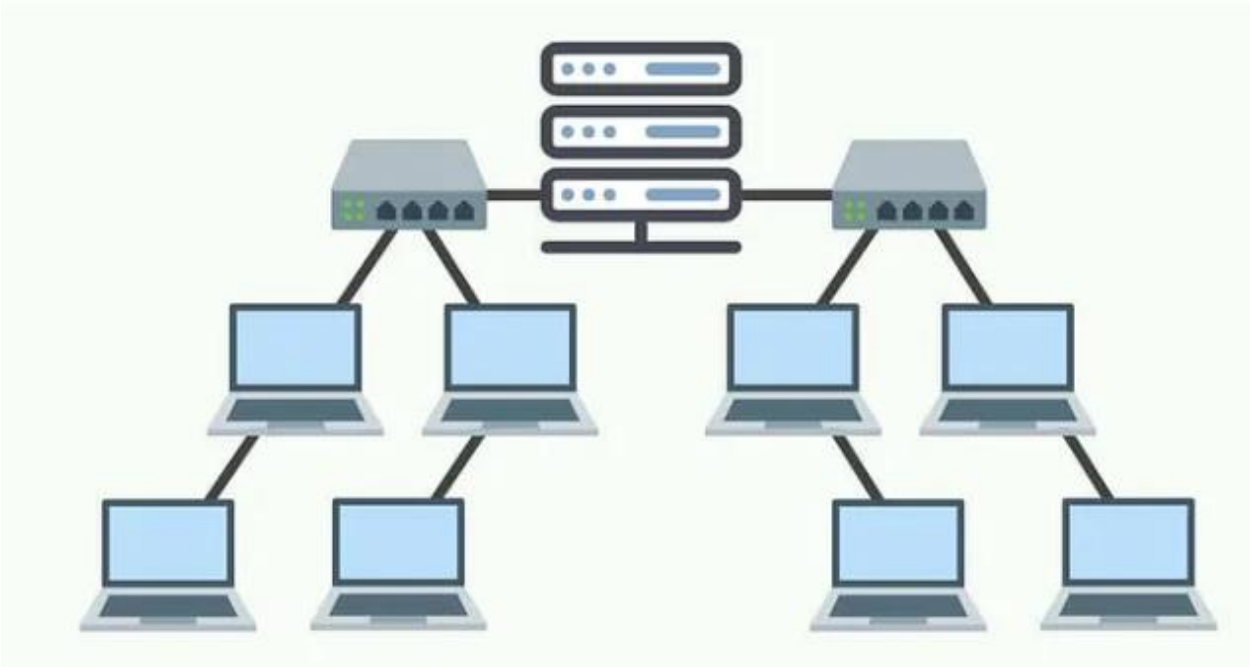


Fig: Tree topology

Advantages:

- It is most suitable for large networks where ring and star topologies are not efficient.
- Since it divides the network in sub-parts, so it becomes more manageable
- There is no disturbance in either expanding or removing the nodes.
- Expansion of Network is possible and easy.

Disadvantages

- The network is weak as it is wholly dependent on the central hub.
- If the network becomes extremely large it becomes difficult to manage.
- Because of its basic structure, tree topology, relies heavily on the main bus cable, if it breakswhole network is crippled.
- As more and more nodes and segments are added, the maintenance becomes difficult.
- Scalability of the network depends on the type of cable used.

HYBRID TOPOLOGY

It uses a combination of any two or more topology. Hybrid networks use a combination of any two ormore topologies in such a way that the resulting network does not display one of the standard topologies(e.g., bus, star, ring, etc.). For example, a tree network connected to a tree network is still a tree networktopology. A hybrid topology is always produced when two different basic network topologies areconnected. Two common examples for Hybrid network are: star ring network and star bus network

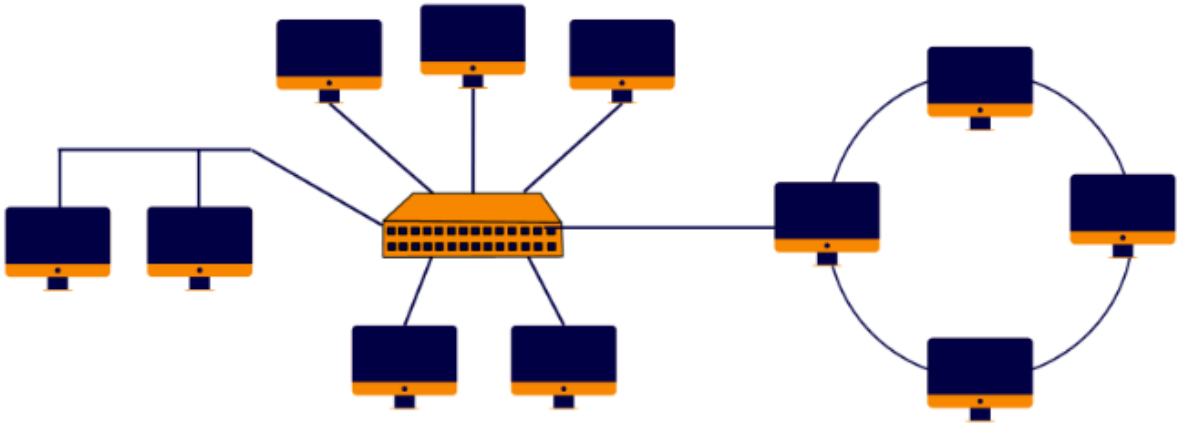


Fig: Hybrid topology

MESH TOPOLOGY

In such structure, all the computers are connected in a network through point-to-point link.

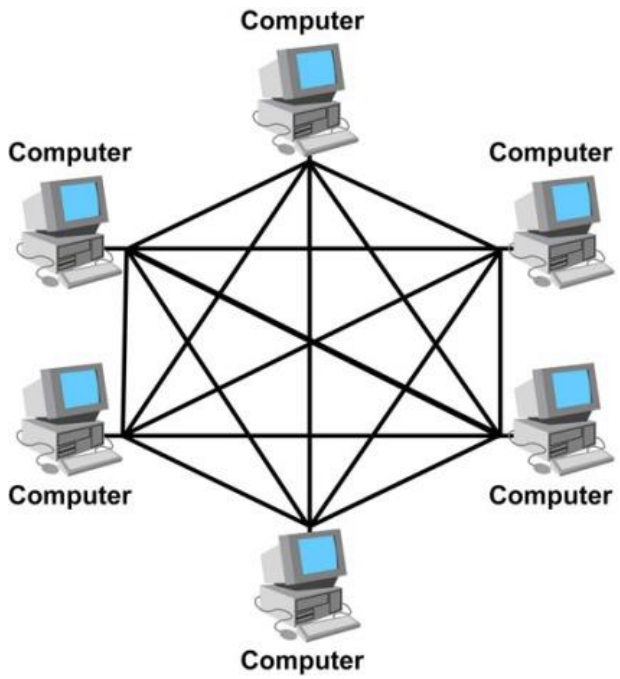


Fig: Mesh topology

Advantage:

- It is very reliable because any link failure will affect only direct communication between the nodesconnected by that link.
- Each node of the network need not have individual routing capability
- Communication is very fast between any two nodes.

Disadvantage:

- It is most expensive network due to point-to-point connection.

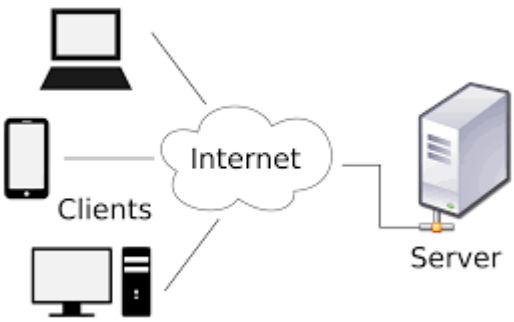
Network Architecture

Network architecture is the design of a communication network. It is a framework for the specification ofa network's physical components and their functional organization and configuration, its operational principles and procedures, as well as data formats used in its operation.

- **Client-Server Architecture**

Client-server architecture (client/server) is a network architecture in which each computer or process on thenetwork is either a client or a server. Servers are

powerful computers or processes dedicated to managing disk drives (file servers), printers (print servers), or network traffic (network servers). Clients are PCs or workstations on which usersrun applications. Clients rely on servers for resources, such as files, devices, and even processing power.

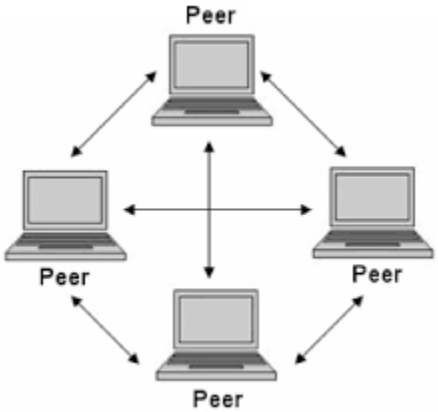


Another type of network architecture is known as a peer-to-peer architecture because each node has equivalent responsibilities. Both client/server and peer-to-peer architectures are widely used, and each has unique advantages and disadvantages.

Client-server architectures are sometimes called two-tier architectures.

- **Peer to Peer Architecture**

Peer-to-peer architecture (P2P architecture) is a commonly used computer networking architecture in which each workstation, or node, has the same capabilities and responsibilities. It is often compared and contrasted to the classic client/server architecture, in which



some computers are dedicated to serving others.

P2P may also be used to refer to a single software program

designed so that each instance of the program may act as both client and server, with the same responsibilities and status. P2P networks have many applications, but the most common is for content distribution. This includes software publication and distribution, content delivery networks, streaming media and peer casting for multicasting streams, which facilitates on-

demand content delivery. Other applications involve science, networking, search and communication networks. Even the U.S. Department of Defense has started researching applications for P2P networks for modern network warfare strategies.

- **Centralized computing Architecture**

Centralized computing is a type of computing architecture where all or most of the processing/computing is performed on a central server. Centralized computing enables the deployment of all of a central server's computing resources, administration and management. The

central server, in turn, is responsible for delivering application logic, processing and providing computing resources (both basic and complex) to theattached client machines.

Workstation

A workstation is a computer intended for individual use that is faster and more capable than a personal computer. It's intended for business or professional use (rather than home or recreational use). Workstations and applications designed for them are used by small engineering companies, architects, graphic designers, and any organization, department, or individual that requires a faster microprocessor, a large amount of random-access memory (RAM), and special features such as high-speed graphics adapters. Historically, the workstation developed technologically about the same time and for the same audience as the UNIX operating system, which is often used as the workstation operating system. Among

the most successful makers of this kind of workstation are Sun Microsystems, Hewlett-Packard, DEC, andIBM.

Internet and E-mail

Internet:

The internet is huge network of computer which links many different kinds of computes all over the world. It is network of network that stores a common mechanism for identifying computers and acommon set of communication protocols for communication between two computers on the network.

The global network of network of networks is called internet. It is considered as the largest information highway. It was created as a project for the US department of defense to transfer data among widely separated computers.

There are basically four types of services provided by the internet to the users.

a) E-mail

The e-mail service allows an internet user to send a mail to another internet user in any part of the world. Messages in e-mail service can contain text documents, images, audio, video, and animateddata.

b) File transfer protocol

It allows internet users to move a file from one computer to another on the internet. A file may contain any type of digital information, text document, image, movie, sound, animation, software's etc. moving file form remote computer to own computer is known as downloading the file and moving a file from own computer to a remote computer is known as uploading the file.

c) Telnet (Remote login)

Telnet service enables an internet user to log in to another computer on the internet from his/her local computer. It is also known as remote login.

d) Usenet news

It allows a group of internet users to exchange their views, ideas, information's on some common topic of interest with all members belonging to the group. Several such groups exist in the internet and are called **newsgroup**. Newsgroup is like a large notice board, which is accessible to all the members belonging to the group. There are two types of newsgroups: **moderated** and **non- moderated**.

- In a ***moderate*** newsgroup only selected members have right to post a message directly on avirtual notice board. In a moderate newsgroup, when an ordinary member creates a new message, s/he first sends it to one of the members having rights to post a

  message on the notice board directly. The authorized member then checks the message to decide its appropriateness for posting. S/he does necessary editing (if any) to improve message's quality and then posts it on the notice board. The message now becomesreadable to other members.

- In ***non-moderate*** newsgroup, any member can posts message on it directly.

Uses of internet:

Internet is a valuable tool for use in many significant way by both commercial and non-profit commercial organizations. Some important applications of internet are:

a) Online communication

It is used as fast and cheaper means of communication. Many users also use chat facility to communicate with each other's in real time. For example: *facebook.com, yahoomessenger.com,* etc

b) Software sharing

The internet provides access to a large number freeware, shareware or even payable software, development tools and utilities through which users can upload and download software. The internet sites *filehippo.com* and *isohunt.com* offers downloadable software.

c) Online shopping

It has facilitated a new market concept consisting of virtual shops. These shops remain opens 24x7 times and are accessible to purchasers all around the world. They provide information about products or services for sale through WWW servers. Using the internet services, customers submitspecific product queries and request specific sales quotes. For example, the internet site *amozon.com* is such a virtual bookshop on the internet, which has information on all types of international books and CDs, and accepts orders for them on-line.

d) Worldwide video conferencing

Using the internet facility, a group of users located around the world can talk and interact witheach other, as if they are sitting and discussing in a single room. Both can interact with each otherby using computer monitor, special audio-devices fixed in their computers

e) On-line journals, magazines, encyclopedia

There are many www sites on the internet dealing with electronics versions of many journals and magazines. There are also sites, which provide access to electronic dictionary or encyclopedia. For

e.g. *Wikipedia.org* provides an online encyclopedia.

f) Organization/product promotion

In this modern era, most organizations create their own web sites to disseminate timely information about product announcement, recent strategies, press release, and other information oftheir products for promotional activities.

g) Customer support service

Many organizations also used the facility of internet to provide timely customer support. For example, users of a software product can report bugs in the software to the organization that owns the software by e-mail and the organization can make available

to its customers bug fixes, known problems and limitations and general advice about the product.

Intranet

The intranet is privately owned organizational network which uses the same protocols as used in the internet to share files and send e-mails. An intranet is cut off from the outside world. It permits its user to access internal file but it prevents outside access to internal files by hiding behind security system known firewall. Using Intranet, organizational policies are secured from introducer (hackers) and other organizations.

The private network structure is intranet. It may exist as LAN or WAN or other form. It is a secured isolated private network structure only by using the "Firewall" techniques. It is a most useful network for sharing the information within the organization. The intranet is a privately owned organizational network which uses the same protocols as used in the internet to share files and send e-mails. An intranet is cut off from the outside world. It permits its user to access internal files but it prevents outside access to internal files by hiding behind a security system known as firewall. Using intranet, organization policies are secured from hackers and other organization.

Advantage of Intranet

- Publishing
- Training
- Workflow
- Document management
- Email
- Better internal communication
- Sharing of resources and best practice
- Improved customer service
- Reduction in paperwork
- Front-end to corporate systems

Extranet

An extranet is a private network securely shares information using same protocols as in intranet with its business partners. An extranet is securely exposed over the internet to specific group that needs to access it. Extranets are powerful tools because they let business partners to share resources on their private network. It is generally used in e-commerce.

An extranet is similar to an intranet but it is made accessible to selected external partners such as business partners, suppliers, key customer for exchanging data and application and sharing information. As with intranet, an extranet can also provide remote to corporate systems for staff that spend lots of time out of the office for instance those in sales or customer support or home workers. Extranet users should be a well-defined group and access must be protected by identification routines and security features. Extranets offer a cheap and efficient way for business to connect with their trading partners can lead to enhanced business relationship and help to integrate your business firmly within their supply chain. An extranet is a private network that securely shares information using same protocols as used in internet with its business partners. An extranet is securely exposed over the internet to specific group that needs to access it. Extranets are powerful tools because they let business partner to share resources on their private network. It is generally used in e-commerce.

Advantage of Extranet

- More integrated supply chains
- Reduced cost
- More effective collaboration between business partners
- Improved business relationship
- Improved customer service
- Single user interface
- Improving the security of communication
- Shared news of product development
- Flexible

e-Mail

The transmission of message that may text, audio, video or combination of these through electronic mediais called e-mail. It enables an internet user to send mail to another internet user in any part of the world in a real time manner. An e-mail message takes a few seconds to several minutes to reach to the destination because it travels from one network to another network, until

it reaches its destination.

Most people use the internet primarily to e-mail and surf the World Wide Web. Simply put e-mail or electronic mail is the processing of sending and receiving message over the internet. The anatomy of e-mail can be explained as let abc@hotmail.com.

The transmission of message that may be text, audio, video or combination of any these through electronic media is called e-mail.

- **Username:** username identifies your account on the email server that handles the e-mail. For e.g. abc located to the left side of @ and then it is the username.
- **@:** It means "at" and separates your account name and the name of the mail server name.
- **Domain name:** The domain name usually consists of two pieces of identifying information. The first piece is the name of the e-mail server and is located to the right of the "@" symbol like "hotmail". The second piece is usually have three letter extension indicates the top level domain. They are separated by periods called "dots" like "com".

Top Level Domains in E-mail Service

Extension	Meaning
.com	A company or Business
.org	A non-profit or not for profit institution
.gov	A governmental Agency
.mil	A military
.edu	A educational institution
.net	ISP or network provider

Types of E-mail

E-mail is one of the most used features of the web. With this technology you can not only send text message but also digital such as image, audio, video etc. attached with the e-mail message. There is little or no cost in sending an e-mail and the best part it's almost instantaneous. There are different types of e-mail as mentioned below:

1. Web based e-mail accounts

All the web-based services provide both webs based free e-mail accounts and paid versions

of the same with extra features. These are called web-based e-mail accounts and services are as hotmail.com, gmail.com, yahoomail.com.

2. E-mail accounts provided by the ISP

ISP generally provides an e-mail account for free with the internet connection. Of you did not get one from your however, there is one major drawback in having your ISP account as your primary e-mail address for online correspondence.

3. Website e-mail accounts

Depending on your hosting package, you can create e-mail on the website for yourself or your staff. Most web hosting companies let you access e-mails.

4. E-mail accounts from your company

If you work in a medium sized or a large company, they would probably have their own e-mail servers. Generally, you get your own e-mail address a few days after joining the company.

Advantage of the E-mail

- The delivery of message is very fast.
- The cost of e-mail is almost free but sometimes negligible communication cost may charge.
- Multiple copies of the same message can be sent to different people.
- High graphics and multimedia file can be attached in e-mail.
- It enhances the communication system.
- It is more reliable and secure than paper mail.
- It is highly secured than paper mail.
- Unlike paper mail, it can be edit using editing program in a computer.
- Time saving.
- Emails are easy to use. We can organize your daily **correspondence**, send and receive electronicmessages and save them on computers.
- Emails are fast. They are **delivered** at once around the world. No other form of writtencommunication is as fast as an email.
- It is possible to send **automated** emails with a certain text. In such a way it is

possible to tell thesender that you are on vacation. These emails are called auto responders.

- Emails do not use paper. They are **environment** friendly and save a lot of trees from being cutdown.
- Emails can also have pictures in them. You can send birthday cards or newsletters as emails.
- Products can be **advertised** with emails. Companies can **reach** a lot of people and inform them ina short time.

Disadvantage of the E-mail

- Although e-mail is delivered on time but receivers may not read their mails on time.
- User should stay online to read and write more e-mail.
- The system is totally based on the internet or other network-based system. So, failure of network may discontinue the e-mail services.
- It is knowledge-based application so all user is not able to use this system.
- Sometimes some unnecessary mails called junks mail may mislead the user.
- Emails may carry viruses. These are small programs that harm your computer system. They canread out your email address book and send themselves to a number of people around the world.
- Many people send unwanted emails to others. These are called spam mails. It takes a lot of time to

  filter out the unwanted emails from those that are really important.
- Emails cannot really be used for official business documents. They may be lost and you cannot

  sign them.
- Our mailbox may get **flooded** with emails after a certain time so you have to empty it from time totime.